Week 7 Project Summary

HealthConnect - Data Science Track

AnalystLab Africa Experience Lab - Week 7 | Data Science Track

1. What I planned and tested

I continued directly from my Week 6 HealthConnect Rwanda Data Science work. I tested the Week 6 hospital-admission candidate model because it is the genuine classification component available in the preserved Week 6 project. The supplied Rwanda data contains no attendance/no-show outcome, and AnalystLab Africa clarified that Week 7 should use the same Week 6 dataset.

2. Most important results

- Week 6 Gradient Boosting reproduction: ROC-AUC 0.4967.
- Week 7 final holdout candidate: ROC-AUC 0.4917, PR-AUC 0.1244, recall 0.0085.
- Refined no-ID balanced Logistic Regression: holdout ROC-AUC 0.5420, but cross-validated ROC-AUC 0.4875 ± 0.0257.
- The refinement was rejected because the apparent holdout improvement was not stable and precision remained too low for operational use.

3. Issues and refinements

The candidate was dominated by false negatives at the 0.50 threshold. I investigated cross-validation, segment performance, feature importance, threshold behaviour and a simplified feature set. The tested refinement increased recall but produced many false positives and failed to demonstrate stable discrimination. No model was promoted to deployment.

4. Cross-track activity

A genuine Data Analytics -> Data Science HC-POD validation was completed with Agu Precious. Her validated findings were used to review feature support and prediction-time leakage. I defined the prediction point as patient arrival / before initial clinical assessment, held Diagnosis and Severity out of the cross-track refinement, recalculated age at appointment, added time-safe prior visits plus appointment month/weekday, and retained FacilityID/DepartmentID with caveats. The cross-track Gradient Boosting re-test produced holdout ROC-AUC 0.5189 and 5-fold CV ROC-AUC 0.5237 +/- 0.0246. This improved feature governance but did not validate a deployable model.

5. Major limitation

The Week 6 Rwanda appointment data has no attendance/no-show/cancellation field. Therefore the overall missed-appointment objective cannot be directly modelled from the supplied data. This is documented as a data limitation.

6. Week 8 readiness

Week 8 should carry forward the validated negative result and completed cross-track feature-governance decisions, request a true attendance outcome if that objective remains central, and collect stronger pre-admission predictors before revisiting deployment.

Overall contribution

My Week 7 contribution is an evidence-based validation of the Week 6 Data Science component. I identified that the current model is not operationally suitable, demonstrated why through reproducible testing, completed a genuine Data Analytics -> Data Science feature validation and re-test, rejected unsupported deployment claims, and defined the next data and integration requirements for Week 8.